

# Coordinated multiple *wh*-sluicing in English: A corpus-based investigation

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## Coordinated multiple *wh*-sluicing (CM-sluicing)

- English coordinated multiple sluicing phenomenon (CM-sluicing), is a type of sluicing in which two fragmental expressions are coordinated (Hankamer and Sag 1976, Chung et al. 1995, Ginzburg and Sag 2000, among others).
  - Someone saw something, but I can't remember **who** and **what**. (Citko and Gračanin-Yuksek 2020: (5))
- It is often compared with other multiple *wh*-constructions such as coordinated multiple *wh*-question (CWHQ) and multiple *wh*-sluicing that is quite restricted in English (Citko and Gračanin-Yuksek 2020).
  - \***Who** and **what** saw? (Citko and Gračanin-Yuksek 2020: (2))
    - \*Someone saw something, but I can't remember **who what**. (Citko and Gračanin-Yuksek 2020: (3))

# Clause-mate conditions

- CM-sluicing is not subject to clause-mate condition which requires all *wh*-remnants to originate in the same finite clause (Merchant 2013), which cannot be violated in multiple sluicing (Vicente 2014).
  - (3) a. [Something was served at the fair [because someone told us to]] but Lee doesn't recall **what** and **who**.  
 b. [[Someone served something at the fair]] but Lee doesn't recall **what** and **who**. (Adapted from Citko and Gračanin-Yuksek 2020: 13)
  - (4) a. \*[Some of the students say [that Harvey talked to some of the professors]], but I don't know **which students to which professors**.  
 b. [[Some of the students talked to some of the professors]], but I don't know **which students to which professors**. (Adapted from Vicente 2014: 51)

## Insensitivity to superiority effects

- The English language is not a multiple *wh*-fronting language, it is under yields superiority effect when two *wh*-phrases undergo movement.
- Since CM-sluicing is not subject to clause-mate conditions, the two *wh*-remnants do not produce a superiority effect (Abels and Dayal 2017a; Kotek and Barros 2018).

- (5) a. Someone talked about something but I can't remember who about what.

(Citko and Gračanin-Yuksek 2020: 17)

b. ..., but I can't remember [CP who<sub>i</sub> [C' about what<sub>j</sub> [TP t<sub>i</sub> talked t<sub>j</sub>]

c. \*..., but I can't remember [CP about what<sub>j</sub> [C' who<sub>i</sub> [TP t<sub>i</sub> talked t<sub>j</sub>]

- (6) a. Someone talked about something but I can't remember who or about what.

(Adapted from Citko and Gračanin-Yuksek 2020: 17)

b. ..., but I can't remember

who [t<sub>i</sub> talked about something] or about what [someone talked about t<sub>j</sub>]

## Restriction on the single-pair reading

- CM-sluicing can be interpreted through the single-pair reading, but not the pair-list reading (Citko and Gračanin-Yuksek 2020).
- Single-pair reading indicates that answers to the CM-sluicing cannot be listed (Abels and Dayal 2017a).
  - (7) a. *Single-pair Reading*
    - b. **Some** student has published on some topic, but I couldn't tell you **which student** and **on which topic**. (Abels and Dayal 2017a: (37a))
    - c. *Possible response(s)*: Kim on coordinated *wh*-sluicing.
  - (8) a. *Pair-list Reading*
    - b. # **Every** student has published on some topic, but I couldn't tell you **which student** and **on which topic**. (Abels and Dayal 2017a: (37b))
    - c. # *Possible response(s)*: Kim on coordinated *wh*-sluicing, Lee on extraposition, Jung on VP ellipsis ...

## Derivation approach

- Originated from Ross (1969) and developed by Merchant (1999), there have been several accounts on multiple *wh*-movements followed by deletion (Kotek and Barros 2018)

(9) a. Sally called someone, but I don't know who.

b. Sally called someone, but I don't know who<sub>i</sub> [<sub>TP</sub> Sally called ~~t<sub>i</sub>~~]

(Kotek and Barros 2018: (2))

- This approach can be applied to CM-sluicing also, where both of the remnants undergo leftward *wh*-movement in separate clauses, for they are not constrained by the clause-mate condition (Citko and Gračanin-Yuksek 2020).

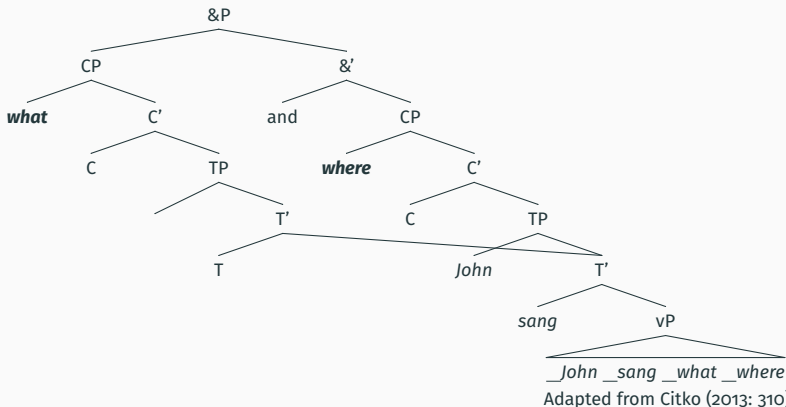
(10) a. Someone saw something, but I can't remember who and what.

b. ..., but I can't remember [<sub>&P</sub> [<sub>CP</sub> who<sub>i</sub> [<sub>TP</sub> t<sub>i</sub> saw something<sub>j</sub>]] and  
[<sub>CP</sub> what<sub>j</sub> [<sub>TP</sub> they<sub>i</sub> saw t<sub>j</sub>]]] (Citko and Gračanin-Yuksek 2020: (54))

## Bi-clausal bulk-sharing structure

- The remnants which originate in the same site are then realized into two different CPs, in which the sluiced elements in bulk are shared by the first remnant, having dominance over the second remnant (Citko 2013).

(11) a. John sang something, but I don't know  
b.

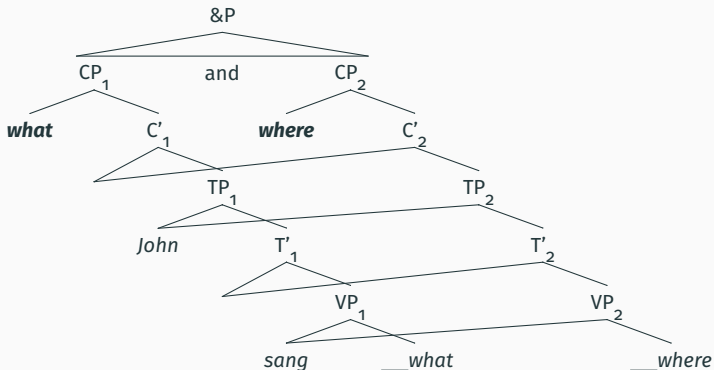


Adapted from Citko (2013: 310)

## Bi-clausal non-bulk-sharing structure

- This structure shows multi-dominance between the two *wh*-phrases on each other, so the structure of the first remnant does not pose dominance on the second remnant (Citko 2013).

(12) a. John sang something, but I don't know



Adapted from Citko (2013: 309)

# LF copying/IP recycling approach

- The LF copying approach assumes that there is an unpronounced element in the sluiced site (Chung et al. 1995; 2010).

(13) a. Joan said she talked to some students, but I don't know who.

b. ..., but I don't know [<sub>CP</sub> who [<sub>IP</sub> she talked to e]] (Spell-Out)

c. ..., but I don't know [<sub>CP</sub> who<sub>i</sub> [<sub>IP</sub> she talked to some students<sub>i</sub>]]  
(LF copying/IP recycling) (Chung et al. 1995: (51))

- This can be applied to CM-sluicing, as in the following:

(14) a. John sang something, but I don't know what and where.

b. ..., but I don't know [<sub>CP</sub> what [<sub>IP</sub> John sang e]] and [<sub>CP</sub> where [<sub>IP</sub> John sang e]]

c. ..., but I don't know [<sub>CP</sub> what<sub>i</sub> [<sub>IP</sub> John sang something<sub>i</sub>]] and [<sub>CP</sub> where [<sub>IP</sub> John sang something<sub>i</sub>]] (Adapted from Citko (2013: 309))

# Research questions

- To observe the distinct characteristics that CM-sluicing have through empirical data.
- To account for the authentic data of CM-sluicing through a Direct Interpretation approach.

## Corpus search

- For this study, COCA (Corpus of Contemporary American English) was used, from all of the nine registers that are available in the corpus.
- The search strings used are:
  - (15) a. \* wh\* and | or wh\* PUNC
  - b. \* how and | or wh\* PUNC
  - c. \* wh\* and | or how PUNC
  - d. \* which NOUN and | or wh\* PUNC
  - e. \* wh\* and | or which NOUN PUNC
  - f. \* which NOUN and | or how PUNC
  - g. \* how and | or which NOUN PUNC
- From over 318 results, a total of 207 tokens were manually sorted for analysis.

# Remnant pairs

- The data consisted of a diverse set of *wh*-remnant pairs, which can be organized in argument-argument, argument-adjunct, adjunct-argument, and adjunct-adjunct pairs.
- (16)
- Someone was keeping something a secret. The only way to get Gella out of there is to find out **what** and **who**. (COCA 1991 FIC) *argument-argument pair*
  - He's moving something with all new guys. I don't want to tell you **what** or **where**. (COCA 2006 FIC) *argument-adjunct pair*
  - I knew I had to do something. I had no idea **why** or **what**. (COCA 1997 FIC) *adjunct-argument pair*
  - Breast cancer can be beaten, they say, and they are here to tell **how** and **why**. (COCA 2002 MAG) *adjunct-adjunct pair*

## Remnant pairs distribution

- The table below shows the distribution of the remnant pairs.

|                   | <i>and</i> | <i>or</i> |
|-------------------|------------|-----------|
| argument-argument | 1          | 4         |
| argument-adjunct  | 17         | 23        |
| adjunct-argument  | 2          | 5         |
| adjunct-adjunct   | 25         | 130       |
| total             | 45         | 162       |

**Table 1:** Remnant pairs by conjunctions

- CM-sluicing prefers the conjunction ‘or’ over ‘and’, and also the ‘adjunct-adjunct’ pair the most out of the four remnant pair types.
- Data with argument-denoting remnants make up 52 tokens of the data, which is about 25% of the 207 data collected.

## Verb type

- This variable was observed to see if there were any relationship between CM-sluing and certain verbs with a simple collostructional analysis, with the `collex` function in R.

| collexeme  | corpus frequency | observation | coll.str.logl | significance |
|------------|------------------|-------------|---------------|--------------|
| know       | 2,762,659        | 102         | 451.12        | * * *        |
| remember   | 298,186          | 12          | 49.09         | * * *        |
| find out   | 93,710           | 8           | 44.20         | * * *        |
| figure out | 56,670           | 6           | 35.62         | * * *        |
| sure       | 453,627          | 11          | 34.41         | * * *        |
| explain    | 193,292          | 5           | 16.16         | * * *        |
| specify    | 15,567           | 2           | 12.60         | * *          |
| ask        | 676,739          | 7           | 11.48         | * *          |
| understand | 349,138          | 4           | 7.19          | *            |
| decide     | 205,460          | 3           | 6.60          | .            |
| tell       | 1,120,504        | 7           | 6.21          | .            |

**Table 2:** Collexeme analysis on verbs of CM-sluing<sup>1</sup>

<sup>1</sup>. = marginally significant ( $0.05 \leq p < 0.1$ ), \* = significant ( $0.01 \leq p < 0.05$ ), \* \* = very significant ( $0.001 \leq p < 0.01$ ), \* \* \* = highly significant ( $p < 0.001$ )

## Other verb types

- Other verbs that were included in the analysis but were not statistically significant enough to mark were:
  - (17) guess, analyze, depend on, investigate, predict, debate, look for, think about, forget, realize, describe, say, show, care, see, have, be
- One thing to note is that CM-sluicing occurs with verbs that require cognitive processes or uttering words.

## Correlate overtness

- This variable was observed to determine whether CM-sluicing preferred sprouting or merger cases - if there were indefinite nouns in the antecedent, it was seen as having an overt correlate, therefore, a merger case.
  - (18) a. Someone was keeping something a secret. The only way to get Gella out of there is to find out **what** and **who**. (COCA 1991 FIC) *overt correlate*
  - b. We need to know more about measurement to decide whether to screen, and if so, **how** and **whom**. (COCA 1991 MOV) *covert correlate*
- The ratio of overt correlates (40 tokens, 77%) to covert correlates (12 tokens, 23%) shows that CM-sluicing prefers to have overt correlates to covert ones.

## Antecedent clause sharing

- This variable marked the relationship between the reconstructed forms of the remnants and their linguistic antecedents, regarding the degree of likeness of their shared form.
  - (19) It was burned down in 1899, but no one seems to know **how** or **why**. (COCA 1996 FIC) *Completely sharing*
  - (20) a. Something had exploded, but none of the men and women could say **what** <had exploded> or **why** <**it** had exploded>. (COCA 2008 FIC) *Partial sharing*
  - b. I always knew that one day I would write and publish a novel, the question was never if but rather **what** <I would write> and **when** <I would write **it**>. (COCA 2012 TV) *Partial predicate sharing*

## Antecedent clause sharing distribution

- The table below shows the two degrees of shared forms, which are either completely or partially sharing.

| <i>and or</i> |    |     |
|---------------|----|-----|
| completely    | 24 | 132 |
| partially     | 21 | 30  |
| total         | 45 | 162 |

**Table 3:** Antecedent source sharing by conjunctions

- CM-sluicing prefers to have their putative source of the remnants to be identical with the antecedent clause.
- Examples that partially share makes up 25% ( 51 tokens ) of the total data, which shows that CM-sluicing prefers completely sharing cases compared to partially sharing cases.

## Coordination of merger and sprouting

- CM-sluicing is able to connect two remnants that are either merger or sprouting cases, which refer to remnants with overt or covert correlates in the antecedent respectively.
  - (21) a. *Someone heard something*, but I don't know **who** and **what**. (Adapted from Citko and Gračanin-Yuksek 2020: (3)) *merger-merger*
  - b. Experts claim that there could be another attack possible, but none of us know **who** and **what**. (COCA 2016 MOV) *sprouting-sprouting*
- Interestingly, it is able to also string remnants with and without overt correlates together.
  - (22) a. What you want to do is prove *something*, but I don't know **what** or **why**. (COCA 2000 FIC) *merger-sprouting*
  - b. *Something* had been achieved, even though he was not quite sure **how** or **what**. (COCA 1994 TV) *sprouting-merger*

## Cases that challenge the derivation analysis

- When there is a sprouting-sprouting case in CM-sluicing, there may not be a site for the remnants to move then elide the redundant elements, as argued from the derivational approach.
  - (23) a. Experts claim that there could be another attack possible, but none of us know **who** and **what**. (COCA 2016 MOV)
  - b. \*Experts claim that there could be another attack possible, but none of us know [[who] [there could be another attack possible] and [what [there could be another attack possible]]]
- This is impossible when there is no gap to derive the ‘who’ and ‘what’ from in the antecedent putative clausal source, hence making move-and-delete process impossible for CM-sluicing in this case.

## Cases that challenge LF copying/IP recycling analysis

- LF copying/IP recycling approach, cannot explain the semantic identity issue if there are two argument remnants that can refer to different entities.
  - (24) a. In this headlong plunge he instinctively reached out and grabbed hold of *something*, he didn't know who or what. (COCA 2011 FIC)
  - b. ? In this headlong plunge he instinctively reached out and grabbed hold of *something*, he didn't know [who [he reached out and grabbed hold of something]] or [what [he reached out and grabbed hold of something]].
- Here, the two remnants are able to share the correlate 'something', but may refer to different entities respectively.
- Even though the syntactic identity can be resolved, it still holds unaccounted issues with the semantic identity.

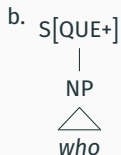
## A Direct Interpretation (DI) approach

- **A Direct Interpretation (DI) approach** assumes that the meanings of the unpronounced material is generated without any syntactic structure. Instead, the sole daughter of an S-node, a fragment, is directly projected to its maximal projection (see Ginzburg and Sag 2000; Culicover and Jackendoff 2005).

(25) **Head-Fragment Construction** (Kim and Nykiel 2020):

Any category can be projected into a NSU (non-sentential utterance) when it functions as a salient utterance (SAL-UTT).

(26) a. Kim met someone, but I don't know who.



# Dialogue and Question Under Discussion

- The interpretation of a sluiced clause depends on the notion of ‘question-under-discussion (QUD)’ in the dialogue, which are lists of possible answers to the current question (Ginzburg and Sag 2000).
- Dialogues are described via a Dialogue Game Board (DGB) where the contextual parameters are ‘anchored’ and where there is a record of who said what to whom, and what or who they were referring to (see Ginzburg 1996, Ginzburg and Fernandez 2010).

## DGB structure

- Uttering the question ‘*Who did Kim meet?*’ will activate the following feature structure with the appropriate DGB information, where each *wh*-question is treated as a proposition in a question (Kim 2015):

(27) Semantic content of *who*:  $\pi^i_{\{\text{person}(i)\}}$

(28) 
$$\left[ \begin{array}{l} \text{FORM } \langle \text{Who did Kim meet?} \rangle \\ \text{SYN } S \\ \text{SEM } \lambda \left\{ \pi^i \right\} \left[ \text{meet}(k, i) \right] \\ \text{DGB } \left[ \begin{array}{l} \text{MAX-QUD } \lambda \left\{ \pi^i \right\} \left[ \text{meet}(k, i) \right] \\ \text{SAL-UTT } \left\{ \left[ \begin{array}{l} \text{SYN } NP \\ \text{SEM } \pi^i \end{array} \right] \right\} \end{array} \right] \end{array} \right]$$

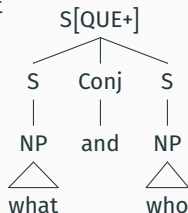
## Argument-argument: Partial sharing

- There is one shared QUD with each *wh*-expression referring to a different argument variable.
- There must be two variables, because each remnant refers to a different entity, as can be inferred from distinct correlates of the antecedent.

(29) a. Someone was keeping something a secret. The only way to get Gella out of there is to find out **what**<sub>y</sub> and **who**<sub>x</sub>. (COCA 1991 FIC)

b. Structure: ... find out [<sub>S</sub> [<sub>NP</sub> what]] and [<sub>S</sub> [<sub>NP</sub> who]]

c. ...find out



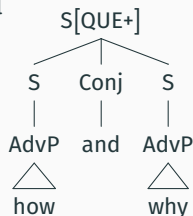
d. QUD:  $\lambda_x \lambda_y [\text{keep}(x, y, \text{secret})]$

## Adjunct-adjunct pairs: Completely sharing

- There is one shared QUD but each sluiced construction is linked to a different variable (manner and reason).

(30) a. Breast cancer can be beaten, they say, and they are here to tell **how** and **why**. (COCA 2002 MAG)

b. Structure: ...tell

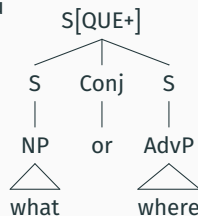


c. QUD:  $\lambda_x \lambda_y [\text{tell}(\text{they}, \text{beat}(\text{we}, \text{cancer}, x[\text{how}], y[\text{why}])))]$

# Argument-adjunct: Completely sharing

- There is only one shared QUD while each sluice is linked to a different variable.
- (31) a. He's moving something with all new guys. I don't want to tell you **what**<sub>x</sub> or **where**. (COCA 2006 FIC)

b. Structure: ...tell you



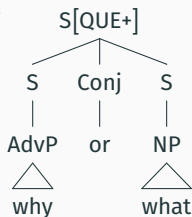
c. QUD:  $\lambda_x \lambda_y [-\text{tell}(I, \text{you}, \text{move}(h, x[\text{something}], \text{with.the.guys}, y[\text{where}])))]$

## Adjunct-argument: Completely sharing

- There is only one shared QUD while each sluiced one refers to a different variable.
- Adjunct-argument pair can be analyzed in the same way as argument-adjunct cases in QUD, even though their ordering is different in SAL-UTT.

(32) a. I knew I had to do something. I had no idea **why** or **what**<sub>x</sub>. (COCA 1997 FIC)

b. Structure: ...no idea



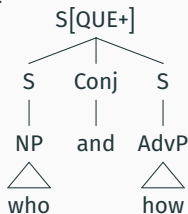
c. QUD:  $\lambda_x \lambda_y [\neg \text{idea}(\text{do}(\text{I}, x[\text{something}], y[\text{why}])))]$

## Voice mismatch

- The DI approach is able to resolve syntactic identity issues such as voice mismatches.

(33) a. And that's all I'm saying, is if you start to look for the why Kennedy was killed, if you start to look for a motive, then you can begin to look for **who**<sub>x</sub> and **how**. (COCA 1991 SPOK)

b. Structure: ... look for



c. QUD:  $\lambda_x \lambda_y [\text{kill}(x, \text{Kennedy}, y[\text{how}])]$

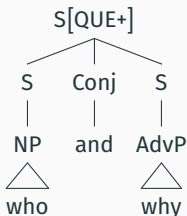
- The QUD evoked here could be 'who killed Kennedy how', in which each sluiced remnants are each linked to a different variable.

## Case mismatch

- Case mismatches, which are also syntactic identity issues, can also be analyzed through the DI approach.

(34) a. You need somebody's help, but I don't know **who**<sub>x</sub> and **why**. (COCA 1990 TV)

b. Structure: ...know



c. QUD:  $\lambda_x \lambda_y [\text{need}(\text{you}, \text{have}(\mathbf{x}, \text{help}), y[\text{why}])]$

# Conclusion

- CM-sluicing is relatively flexible construction that is able to accommodate various structures as its antecedent.
- The distribution of CM-sluicing in COCA shows that it can be divided into two subtypes: examples that completely or partially share.
- Each clause of the CM-sluicing shared one QUD, while each sluiced phrase is linked to a different variable.
- The DI approach requires no complex syntactic structure for the semantic resolution. This appears to be much a simpler account for the CM-sluicing examples.

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